Title:
Development of a Local AI Assistant for Automatic Metadata Extraction and Semantic Search in Scientific Data Management Systems

Research question:
To what extent can local, open-source LLMs automate metadata extraction and semantic retrieval over heterogeneous scientific data, and how do they compare to cloud-based APIs in accuracy, latency and cost?

Abstract:
Modern research laboratories generate large amounts of heterogeneous digital information, including experimental reports, machine output files, images, CSV files, PDFs and analysis results. A major challenge is that valuable scientific information is often stored in unstructured files, making it difficult to organize, search and reuse. The objective of this Master Thesis is to design and implement an AI-based system capable of automatically extracting relevant metadata from scientific files and enabling intelligent search over experimental data.

The project will be developed in the context of FAIR-LAB at IMDEA Materials, a scientific data management platform focused on applying FAIR principles (Findable, Accessible, Interoperable and Reusable) to materials science workflows such as additive manufacturing and X-ray computed tomography.

Problem definition:
Researchers currently spend significant time manually reading experimental files, extracting important parameters and organizing information. The goal is to automate part of this process by combining document processing, Large Language Models (LLMs), vector databases and Retrieval Augmented Generation (RAG). To keep experimental data inside IMDEA and avoid external API costs, the system is built on open-source local models. The student will build a prototype pipeline following this workflow:

Scientific files
(PDF certificates, CSV, logs, reports)
|
v
Metadata extraction pipeline
(Python, OCR, parsers)
|
v
Embeddings
(local embedding model)
|
v
Vector database
(FAISS / Chroma / pgvector)
|
v

RAG system
|
v
Local LLM
(Ollama: Llama, Mistral, Qwen, etc.)

Scope (core + extensions):
To ensure a deliverable result within the duration of the thesis, the scope is structured as a minimal core plus optional extensions.

Core (must deliver):
- Metadata extraction from one primary file type used in FAIR-LAB workflows.
- Structured metadata generation and storage.
- Semantic search using embeddings stored in pgvector (PostgreSQL), reusing the existing FAIR-LAB database stack.

Extensions (if time allows):
- A RAG assistant for natural-language querying over the data.
- Benchmarking local models against each other and, optionally, against a cloud API.
- Support for additional file types.

Main tasks:
1. Analyze scientific data formats used in FAIR-LAB workflows (PDF reports, machine logs, CSV files and metadata records).
2. Develop automatic extraction pipelines to convert unstructured information into structured metadata.
3. Implement a semantic search system using embeddings and a vector database (pgvector by default, within the existing PostgreSQL stack).
4. Develop a Retrieval Augmented Generation (RAG) assistant that allows users to query scientific data using natural language.
5. Evaluate the system by measuring extraction quality, retrieval accuracy and response usefulness against a small manually annotated gold set (approx. 30–50 files with known correct metadata), using a simple human-evaluation rubric for response quality.

Technologies:
- Python
- Artificial Intelligence / Machine Learning
- Large Language Models (LLMs)
- Local open-source models served via Ollama (Llama, Mistral, Qwen or similar)
- Retrieval Augmented Generation (RAG)
- Vector databases (pgvector by default; FAISS or ChromaDB as alternatives)
- SQL databases (PostgreSQL)
- APIs
- Docker

Available resources:
The project does not require training a large AI model from scratch or large amounts of labelled data. Existing FAIR-LAB datasets, metadata structures, experimental documents and public scientific resources will be used to develop and evaluate the prototype. The system runs open-source local AI models inside IMDEA infrastructure, avoiding dependency on external paid AI services and reducing token costs to zero. External AI APIs may only be used optionally for comparison.

Expected outcomes:
- Prototype AI assistant for scientific data management.
- Automatic metadata extraction workflow.
- Semantic search engine for experimental datasets.
- Evaluation of local AI models for scientific information retrieval, including an accuracy/cost comparison.
- Documentation and recommendations for integration into FAIR-LAB.

This project combines artificial intelligence, data engineering and software development, providing experience in building AI tools for next-generation scientific research infrastructures.